



GROUNDWATER INTELLIGENCE

AI-POWERED MONITORING, ANOMALY DETECTION & DEPLETION FORECASTING

Powered by:

EXASOL | Python | DBeaver | Excel | PowerShell

Team:ASAP

GSSS Institute of Engineering and Technology for Women

THE PROBLEM: INDIA'S GROUNDWATER CRISIS

The Problem

- India depends heavily on groundwater for agriculture, drinking water and livelihoods. Continuous depletion can push regions into Critical or Over-Exploited conditions.

What Already Exists

- Thousands of Digital Water Level Recorders (DWLRs) monitor groundwater levels.
- India-WRIS provides water-resource information.
- CGWB assesses groundwater conditions and classifies assessment units.

Our Goal

- Transform groundwater data into predictive, actionable intelligence before depletion becomes a crisis.

THE DATA EXISTS. THE INTELLIGENCE GAP DOESN'T HAVE TO.

Existing Ecosystem

DWLR Sensors



Groundwater observations



India-WRIS / Water-Resource Data



Monitoring & Reporting

Current limitation:

Current approach

- Record water levels
- Observe anomalies
- Identify current depletion
- React after crisis

Our required approach

- Interpret trends
- Detect sensor faults
- Forecast future risks
- Enable early action

Our Core Insight

The challenge is not lack of data.

It is the lack of an integrated system that converts:

DATA → CLEAN DATA → PATTERNS → FORECAST → RISK → EARLY WARNING

Our Proposition

Build an AI-powered intelligence layer on existing groundwater monitoring infrastructure.

OUR PROPOSED SOLUTION

AI-Powered Groundwater Intelligence Platform

1. DATA INGESTION

- DWLR observations
- Historical groundwater levels
- Rainfall data
- Seasonal patterns
- Station / block / district information

2. DATA PREPARATION

Cleaning → Missing-value handling →
Standardization → Quality checks

3. INTELLIGENCE ENGINE

Anomaly Detection

Identify suspicious sensor readings

Trend Analysis

Identify persistent groundwater decline

Forecasting

Predict future groundwater conditions

4. RISK CLASSIFICATION

Normal → Watch → High Risk →
Critical

5. ACTIONABLE DASHBOARD

- Current condition
- Historical trend
- Forecast
- Risk score
- Sensor status
- Early-warning alerts

DATA & TECHNOLOGY STACK

How We Built the Project

DATA SOURCES

CGWB

- Groundwater assessments
- Groundwater classification
- Monitoring information

India-WRIS / NWIC

- Water-resource and groundwater information

Rainfall & Seasonal Data

- Supporting variables for trend and forecasting analysis

OUR TECHNOLOGY WORKFLOW

EXCEL

Data inspection & initial preparation



DBEAVER

Database connection, SQL queries & data management



EXASOL

High-performance analytics & large-scale processing



PYCHARM

Python development & AI/ML workflow



POWERSHELL

Terminal-based execution & project workflow

THE INTELLIGENCE ENGINE - DETECT → UNDERSTAND → PREDICT

1. SENSOR ANOMALY DETECTION

The system analyses:

- Historical station behaviour
- Sudden level changes
- Missing/inconsistent observations
- Temporal patterns

Output:

Normal Reading / Potential Sensor Anomaly

2. GROUNDWATER TREND ANALYSIS

Identifies:

- Persistent water-level decline
- Rate of decline
- Seasonal variations
- Changes from historical behaviour

Temporary fluctuation ≠ Long-term depletion

3. DEPLETION FORECASTING

Historical Levels + Seasonality + Rainfall + Trends

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Forecast Model

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Future Groundwater Condition

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Risk Level

Normal | Watch | High | Critical

FROM AI OUTPUT TO EARLY ACTION

GROUNDWATER INTELLIGENCE DASHBOARD

Dashboard Indicators

CURRENT STATUS

- Current groundwater level
- Latest reliable observation
- Station status

TREND

- Rising / Stable / Declining
- Rate of change

DATA QUALITY

- Normal
- Suspicious
- Potential anomaly

FORECAST

- Future groundwater trend
- Predicted risk

ALERT

- Areas approaching critical thresholds

WHO BENEFITS?

District Authorities

- Identify high-risk blocks
- Prioritize interventions

Water-Resource Departments

- Regional planning
- Resource management

Farmers

- Understand local groundwater stress
- Receive simplified risk information

DECISION LOOP

SENSORS → DATA → EXASOL → AI → RISK SCORE →
DASHBOARD → EARLY ACTION

IMPACT, SCALABILITY & FUTURE

Building a Predictive Groundwater Ecosystem

KEY IMPACT

- Earlier Detection

Identify emerging depletion sooner.

- Reliable Data

Flag faulty or suspicious sensor readings.

- Risk-Based Planning

Focus resources on high-risk regions.

- Scalable Analytics

Analyse large monitoring networks using Exasol.

- Better Decisions

Convert raw data into forecasts, risk scores and alerts.

SCALABILITY

Pilot Districts

Regional Deployment

National Groundwater Intelligence Network

FUTURE DATA INTEGRATION

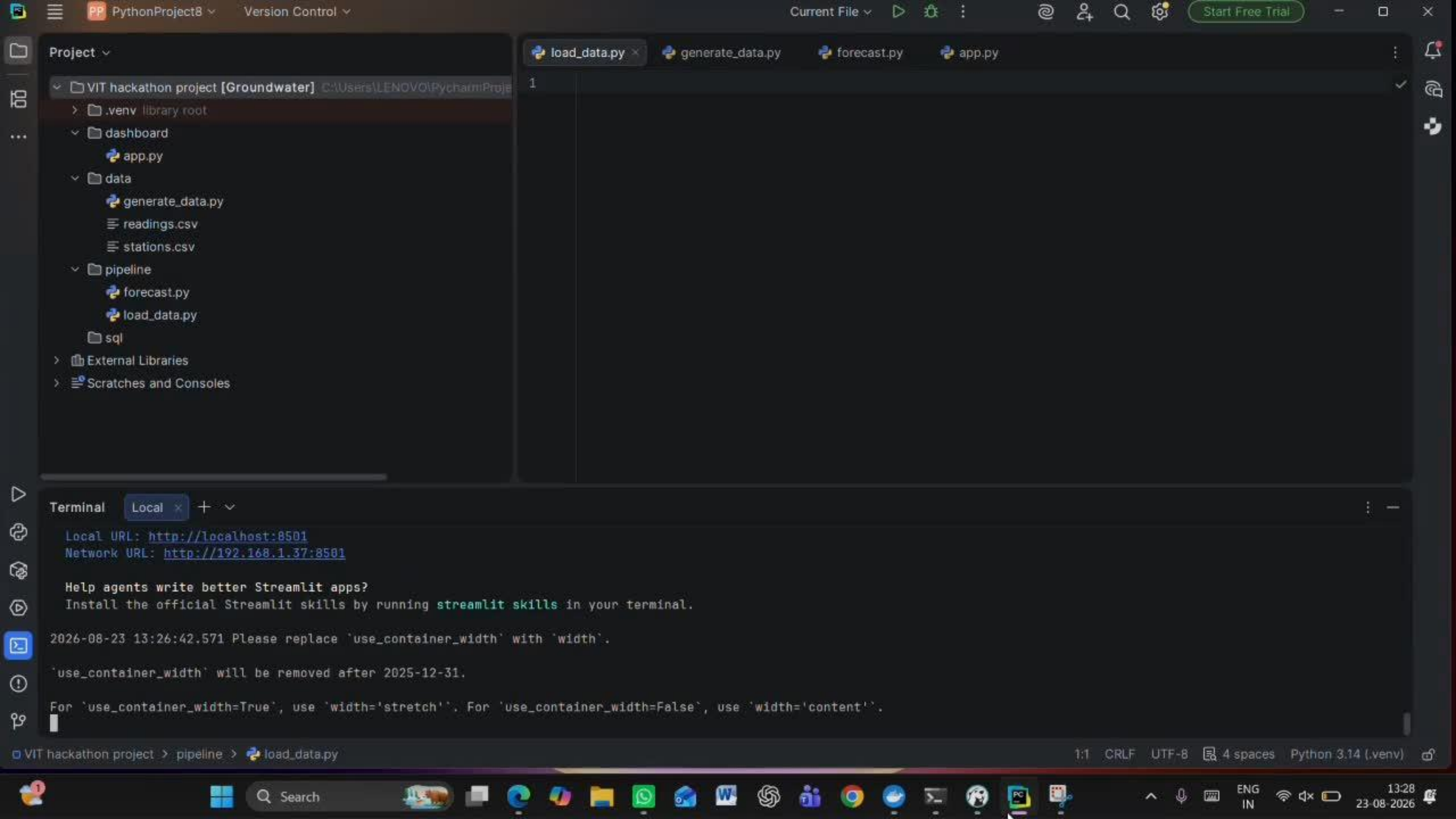
- More DWLR stations
- Rainfall
- Soil & land-use information
- Agricultural water demand
- Recharge & extraction data
- Additional environmental variables

DEPLOYMENT POTENTIAL

Government → Decision-support platform

Water Organizations → Risk analytics & planning

Farmers → Localized groundwater alerts





THANK YOU